Step 8 summary - final mock CSC662 paper vs 3 clean modern papers

Scope. mock_paper_04.pdf (6 pages), built from the final hardened realistic run seed 12 (DeepSeek V4.1 Flash), vs JUN 2019 (7), DEC 2019 (6), FEB 2022 (14). Word COM export; 200 dpi renders. Full report: step8_comparison.md. Paper-01 version archived as step8_comparison_paper01.md.

Structure

Feature	Mock 04	JUN 2019	DEC 2019	FEB 2022
Part A / B	10 x (2+6) = 80 / 2 x (5+5) = 20	same	same	same
Total / duration	100 / 3 HOURS	100 / 3 HOURS	100 / 3 HOURS	100 / 3 HOURS
Part headings	PART A (80 MARKS) / PART B (20 MARKS)	same	same	same
Cover	FINAL EXAMINATION, 5 instructions	same style	same style	FINAL TEST, 3 instructions + name/marks grid
Printed pages	6	7	6	14
Computed / theory marks	6 / 94, 0 placeholders	6 / 94	6 / 94	0 / 100

Matches: parts, question counts, 2/6/5/5 marks, 100 total, 3 hours, a/b numbering, part mark totals, header/footer on every body page, right-aligned mark labels. **Differences:** 6 vs 7/14 pages (matches DEC); cover differs from FEB 2022; computed share differs from FEB.

Layout (pages 2+)

	Mock 04	JUN 2019	DEC 2019	FEB 2022
Header / footer present	5/5 / 5/5	6/6 / 6/6	5/5 / 5/5	13/13 / 13/13
Header / footer / body ink (median)	0.0240 / 0.0070 / 0.0428	0.0038 / 0.0102 / 0.0368	0.0116 / 0.0105 / 0.0459	0.0125 / 0.0166 / 0.0184
Question heading	10.7 pt bold, x=0.123	OCR only	9.7-10.0 regular, x=0.136	11.0 bold, x=0.121
a/b labels / marks labels	24 / 24	24 / 24 (OCR)	24 / 24	25 / 24
Mark right edge / page width	0.879	OCR only	0.868	0.879

Body ink is now close to DEC (0.0428 vs 0.0459), so the 6-page paper matches DEC density. Header ink is heavier than all three real papers (bold header); footer ink is lighter. FEB 2022 gives each question a page; the mock is tighter and has less answer space.

Computed share

Mock: **6 computed marks (1 Vigenere slot) = 6 %**, 94 theory marks across 23 sourced questions. Clean modern fingerprint: **40 machine-verifiable marks over 10 papers = 4.0 marks/paper**; the three comparison papers carry 6, 6 and 0. The paper is realistic, not practice mode, and supersedes paper 01's 28 computed marks / 28 %. Only one cipher type, so no repeated Caesar/Playfair slots.

Theory content and wording

23/23 theory questions accepted; 94 marking points; 0 placeholders; source overlap 0; max near-copy **0.614** (threshold 0.80). Low stem-overlap: **36/94 (38.3%)**; entailment initial yes 22 / partly 12 / no 2, both no points regenerated; final yes 23 / partly 13 / no 0. Verbatim-copy rejections: 6. Both Part B groups have a shared scenario (grounding 0.391), versus the measured 7/19 = 36.8 % rate.

Marks	Mock median words (range)	Clean modern median (p90)
2	20 (18-30)	9 (18)
5	31.5 (24-36)	26 (43)
6	31 (28-45)	18 (44)

Generated 2-mark stems are the main wording gap (20 vs 9 words); 6-mark stems are also longer (31 vs 18). Command mix this seed: Explain 11, State 6, Describe 2, Discuss 2, List 2; no Compare/Propose/Analyse/Justify. Closest past pair: A1a ethics vs CSC662-2018-06 Q15a = 0.614. The stems are grounded and non-copied, but wordier and more scaffolded than real questions.

Marking scheme

mock_scheme_04.pdf, 15 pages, same header/footer as the paper: computed Vigenere working + 26 x 26 tableau, every theory point with verbatim quote and slide citation, *supported by lecture slides (not lecturer-marked)* label, stem-overlap scores, low-overlap flags and supported/partly-supported entailment verdicts. No official CSC662 schemes exist; this is far more detailed than a real scheme.

Weakness list (specific)

- 2-mark stems too long: 20 vs 9 words median; 6-mark 31 vs 18.
- Command mix narrow in this seed: no Compare/Propose/Analyse/Justify.
- Header ink 0.024 vs 0.012 real (bold running header); footer lighter (0.007 vs 0.010-0.017).
- Body density close to DEC but much denser than FEB; 6 vs 14 pages, less answer space.
- 36/94 marking points (38.3%) remain low stem-overlap; accepted by entailment but flagged for review.
- Both Part B groups have scenarios (over-represents the 36.8 % measured rate).
- No official marking scheme; 15-page scheme is far more detailed than real.
- Theory answers are slide-supported, not lecturer-marked.
- Computed share 6 % vs 4.0-mark modern average (within range but 1.5x).
- Cover matches JUN/DEC but differs from FEB 2022.